

Please provide your responses to the following questions. Point values indicated the expected depth of response.

1. Describe some of the beneficial safety features of SFRs. (6 pts)
2. Why do we alloy uranium for fuels? (6 pts) - α -U, anisotropy of
3. Why must metallic fuel systems include a relatively low smear density and large plenum? (10 pts) write a bit more? Sodium - lumps? T_K
4. What is constituent redistribution in metallic fuels? Why does it occur? What are its impacts? (14 pts) thermal gradient, solubility gradient,
5. How does the thermal conductivity vary as a function of burnup in metallic fuels? What phenomena drive this behavior? (8 pts) less U, more Pu?, more FP
6. For metallic fuel: What is FCCI? What are the adverse effects of FCCI? What are the primary fuel and cladding species participating in FCCI? (14 pts) rare earths - Nb, Mo, Tc, Rb, Rh, Pd
7. Describe restructuring in MOX fuels, including why it happens. (12 pts)
8. How does Pu and O concentration vary spatially in a MOX pin? (10 pts)
9. What is JOG? Why does it occur? What are its impacts? (12 pts) Cs, Te, I, Mo Ba
10. Discuss the means which sodium corrosion occurs in SFRs. (8 pts)

related?

more, also check species

2) isotropic center

α -U anisotropic outer cooler region

evap - where gas liquid

Na not corrosive but FP and impurities can lead to corrosion (Hydrides?)

interstitial defect leading to void loop on {010} planes

void leads to loop on {110} planes

this is bad and can be prevented by alloying U.

$\sigma_{radial} > \sigma_{axial}$ by 2x due to anisotropy.

JOG, hypostoichiometric, O/U ratio 1.9 instead of 2.0

Pu moves up thermal gradient leaving gaps in the intermed. region

U moves down th. grad. moving the gaps in intermed. region

< 500 GB practically in 500-600

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1) SFRs ~~can operate~~ use Na ~~as a~~ Na as coolant. Na has high thermal conductivity, melts at ~~too~~ relatively low temps (371 K) and boils at 1156 K giving a wide ~~range~~ operating temp range for Reactor. These melt and boil temps are at atmospheric pressure so the Reactor pressure vessel does not need to ~~be~~ handle high pressures such as PWRs.

Na also ~~has~~ is a weak neutron moderator which is good for fast Reactors.

~~The~~ The high thermal conductivity of Na and its wide operating temp mean that in a worst case situation like LOFWS (Loss of Flow w/ Station Blockout and ~~no~~ no operator input was tested!) ~~It did not~~ The EBR II did not even SCRAM.

~~The~~ The SFRs have passive safety. ~~As temps~~ SFRs also allow for overpower transient events and load following.

The SFR ~~is~~ is inherently safe since as Na temps go up (it is still below boiling temp), the reactor core components will thermally expand, ~~the~~ which then causes a higher neutron leakage in the core leading to a slowdown of the fission chain reaction. As the cycle continues (due to loss of flow) the fission chain reaction will eventually slow to a stop.

2) ~~We~~ alloy U for fuels

3) Metallic fuel system must use a low sinter density ($\sim 75\%$) and a large fission gas plenum ($\sim 1-1.5 \times$ fuel by vol) to prevent the massive amount of swelling the earlier ~~more~~ ~~old~~ fuels ~~had~~ - this was due to the early fuels being more

- all have!

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~~denser~~ ^{? - no} dense. The lower sinter density essentially makes the fuel less dense and has pores that allow the fission gas to escape. The fission gas plenum is from the fuel. Once the gas escapes, it can ~~go~~ travel thru the plenum into filters, ~~and what~~. Plenum also allows off-gassing ^{gas trapped in plenum in cladding tubes}. Prior to the low sinter density and plenum, ~~it~~ ^{confused} metallic fuels would see swelling of ~~upto~~ ^{~ 30%}.

2) We alloy Uranium for 2) We alloy Uranium for fuel for better material and engineering properties. There's α -U swelling, ~~is~~ where in the hotter fuel centerline, the center is isotropic. The α -U on the cooler periphery is anisotropic. The anisotropy is from interstitial defects leading to loops on $\{010\}$ planes and void ~~loop~~ ^{forming} into loops on $\{110\}$ planes. This anisotropy leads to a 2x higher radial stress than axial stress. Alloying Uranium helps mitigate this. - δ -phase stability

4) W/ continued redistribution in metallic fuels, Pu ^{$\rightarrow Zr$} moves up the thermal gradient leaving gaps in the intermediate region. U moves down the thermal gradient into the gaps in the intermediate region left by Pu. This occurs due to the burnup? in metallic fuels as well as how different constituents move across the solubility gradient. There are also constituent redistribution also occurs due to Fission Products (FP) from the fission process. Basically, as the fission process happens, constituent redistribution will take place. ^{nope} This redistribution can lead to FCM (Fuel cladding mechanical interaction), FCM (Fuel cladding chemical interaction), PCM (Pellet cladding mechanical interaction), ~~etc~~, the PCM and FCM lead to structural ~~not~~ stresses on the fuel ~~rod~~.

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- nope. didn't really get it...

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and may lead to swelling. The FCCI would lead to ~~corrosion~~ corrosion.

6) ~~FCCI~~ ^{7/14} FCCI (Fuel-Cladding chemical interaction) is when ~~the~~ ~~FPS~~ ~~from~~ fission products (FPs) ~~from~~ ~~from~~ the fuel ~~interact~~ ^{interact w/ each other to} chemically interact w/ the cladding. When the ~~reacts~~ fuel and cladding are mechanically touching, the ~~elements~~ ~~fuel~~ ~~at~~ ~~FP~~ ~~a~~ different species can ~~interact~~ ^{interact w/ each other to} mutate and cause damage, change the mechanical & thermal properties of the fuel and cladding and can lead to corrosion. The primary fuel and cladding species are Mo, Tc, Rb, Rh, Pd, and rare earths such as Nd. FCCI is a problem also here.

- not really
- FP into fuel, U/Pu into clad
- low T_{melt} phase, clad wastage

7) For MOX ^{restructuring} ~~redistribution~~ there are 5 solid solutions that are metallic: Mo, Tc, Rb, Rh, Pd. There are 3 ^{species} that ~~go into~~ have high solubility like Cs, Te, I. There is a "gray-state" ~~matter~~ ^{that species} Ba, that makes an oxide. ~~not FP states~~ The restructuring takes place based on thermal gradient, solubility gradient, and fission products (FPs). The Mo in the SS cladding gets replaced by another species, and the chromium layer. The chromium layer in the SS can pit when. Below $500^{\circ}C$ the GB plastically deforms to handle the strain due to strain leading to cavitation and tearing at the ~~GBs~~ GBs (grain boundaries). Between $500-600^{\circ}C$ the fuel undergoes restructuring probably due to ~~turning~~ ^{turning} where the GBs ~~to~~ ^{exposed} structure or grains undergo turning. ~~This~~

got your concepts mixed here. Seems like you're discussing 2-4?

8) ~~totally~~ In a MOX pin, ~~you have~~ we have as sintered on the colder periphery, equiaxed region, columnar region, and the void center (void is the hottest). ~~As the~~ The plutonium and oxygen concentration needs to vary uniformly ~~X~~ in the fuel during fabrication. Otherwise,

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this discussion I wanted for #7

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-this happens!

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the fuel will get localized hot spots where the temperature is considerably higher and thermal conductivity is lower (locally).

7/12 9) JOG (Joint Oxide Growth) is where the protective chromia layer has excess oxygen and when the Mo from the chromia layer moves it leaves pitting and behind and allows the corrosive FPs to damage the cladding usually. But there is a "rind" due to the

The rind that forms is the JOG, it protects the general protective layer that prevents the FPs from corrosive fission products from damaging the cladding. To prevent the drawbacks of JOG can be fixed w/ a hypostoichiometric ratio where the O/M ratio is 1.9 instead of 2. It allows higher thermal conductivity. One of its drawbacks is that it can put radial loads on the fuel cladding. - some pieces correct, some confused

5/8 10) Na is generally not corrosive to components but traps FPs and other impurities can end up corroding components and various systems in SFRs. Na purity levels must be kept in check, otherwise oxides, hydrides, nitrides or other impurities will cause accelerated wear and tear.

- thermal mass transport? solubility?

6) Additional info: The protective chromia layer Mo in SS cladding can leave the chromia layer (chromia is the protective layer) and have the cladding be subject to the FPs and other species participating in FCCI.

5) Additional info: As more FPs are off-gassed over time and fuel gets ^{swollen} ~30% of

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~~initial volume~~ higher of initial volume, the pores

5) AB with higher burnup, the ~~FP~~ FP and fission gases off gas from the pores of the fuel into the plenum. This takes place also. When this takes place the metal fuel also swells. The ~~are~~ The pores in the fuel get ~~logged w/ sodium~~ filled w/ sodium drastically increasing the thermal conductivity. While the sodium has not filled the fuel, it ~~will~~ ~~subject~~ and it is still releasing gas, the thermal conductivity will be a little lower due to the gases.

Phenomenon called "sodium logging".

7/8

- Swelling is from gases, swelling stops on FG release
- most of it